

1. What is caching?

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| ☐ An in-memory database | ☐ A high-speed data storage layer |
| ☐ A global network for content distribution | ☐ A way to store passwords |

2. Which types of data should you cache?

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| ☐ Specialised data that is needed by a subnet of users | ☐ Data that can be retrieved quickly with simple queries |
| ☐ Static data that is frequently accessed | ☐ Dynamic generated web content |

3. What is a benefit of caching?

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| ☐ Load balancing the application | ☐ Reduced response latency |
| ☐ Increase application reliability | ☐ Decreased costs |

4. What does Amazon CloudFront enable?

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| ☐ Bidirectional caching between users and an origin host | ☐ Transactional processing with an in-memory database |
| ☐ Automatic creation of a time-to-live value | ☐ Multi-tiered and regional caching of content |

5. How does Amazon CloudFront use edge locations?

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| ☐ It caches local content at edge location. it deliver the cached content to clients through the edge location that requires the fewest network hops to reach those clients. | ☐ It caches all content from an origin distribution at the edge location, and delivers the content to clients through the fast edge location. |
| ☐ It caches Regional data at Regional edge locations, and delivers the content to clients through their Regional edge locations. | ☐ It caches frequently accessed content at edge locations. It delivers the cached content to clients through the edge location with the lowest latency to those clients. |

6. Where is the application session data cached when using sticky sessions?

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| ☐ Elastic Load Balancing load balancer | ☐ Web browser |
| ☐ Web server | ☐ Amazon CloudFront |

7. Which statement best describes an efficient way to deliver on-demand steaming content by using Amazon CloudFront?

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| ☐ A best practice is to create video segments and store them in an Amazon S3 bucket. Then, use CloudFront to cache the segments. | ☐ A best practice is to create distributions for each Region where you serve streaming content. |
| ☐ CloudFront does not work with steaming content. | ☐ A best practice is to create separate origin server for each Region where you serve streaming content. |

8. What is Amazon DynamoDB Accelerator (DAX)?

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| ☐ A feature of DynamoDB that automatically adjusts read/write capacity to handle load | ☐ A feature of DynamoDB that enables fast lookup of items by using secondary keys |
| ☐ A fully managed, highly available, in-memory cache for DynamoDB | ☐ A fully managed, highly available cache that is backed by DynamoDB |

9. How can an application use Amazon ElastiCache to improve database read performance? (Select TWO.)

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| ☐ Read data from the database first and write the most frequently read data to ElastiCache. | ☐ Replicate the database in ElastiCache, and direct all reads to ElastiCache and all writes to the database. |
| ☐ Write data to ElastiCache whenever the application writes to the database. | ☐ Read data from ElastiCache first and write to ElastiCache when a cache miss occurs. |

10. Which role does Amazon CloudFront play in protecting against distributed denial of service (DDOS) attacks?

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| ☐ Performs deep packet inspection to detect attacks | ☐ Route traffic through edge locations |
| ☐ Control traffic by the source IP addresses of requests | ☐ Restricts traffic by geography to help block attacks that originate from specific countries |

